

ELEC70122: Machine Learning For Safety-Critical Decision-Making

Lecture 7: Causal Inference: Part 2 – Effects of Causes

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Credit: Several slides have been modified from Sontag 2019

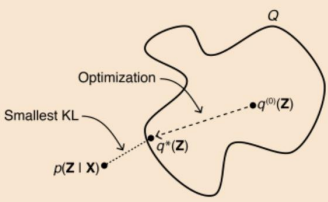
Recap from last lecture – Learning Causes of Effects

A Modelling functional relationships (causal mechanisms)

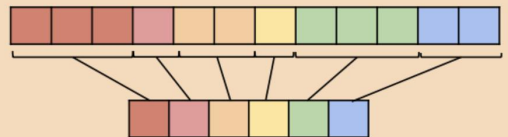
$$X_i = f_i(X_{pa(i, \mathbf{G})}, U_i)$$

$X_{pa(i, \mathbf{G})} \rightarrow \text{NN} \rightarrow X_i$

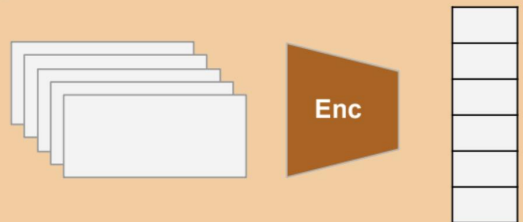
B Learn a distribution over graphs



C Learn representations of causal variables as rich compositions of learned features



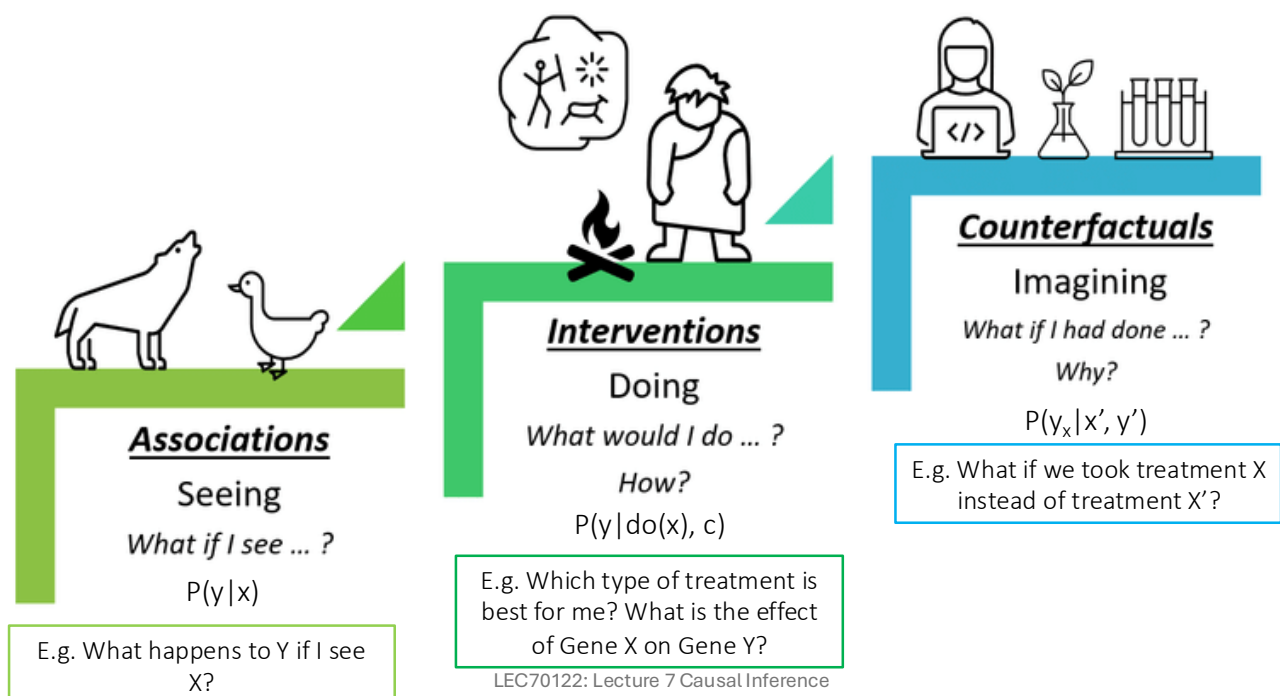
D Learn latent causal variables



Outline

- Potential Outcomes
- Average Treatment Effect (ATE) and Conditional ATE (CATE)
- Ignorability
- Algorithms for estimating ATE and CATE
 - Covariate Adjustment and Propensity Scores
 - Neural Network Approaches to Treatment Effect Estimation

Pearl's Causal Hierarchy



Common Frameworks for Causality

- Potential Outcomes (Rubin-Neyman)
- Causal Graphs (Judea Pearl)
- And several others

These frameworks all suffer from **the fundamental problem of causal inference**: it is impossible to observe both outcomes (treatment and control) for the same individual at the same time.

Each framework thus requires manipulating reality.

Potential Outcomes

- Unit: a person, bacteria, company, school, family, website etc
- Treatments/Actions/Interventions (**A/B**)
- Potential Outcomes:
 - Y_1 : the unit's outcome had they been subjected to treatment $T=1$
 - Y_0 : the unit's outcome had they been subjected to treatment $T=0$.
 - If the number of treatments is T , then there are T potential outcomes (T possibly infinite)
- A single unit gets one of T treatments

Assumptions for Potential Outcomes Framework

- **Stable Unit Treatment Value Assumption**

The potential outcomes for any unit do not vary with the treatments assigned to other units

Failure cases: vaccination, network effects

For each unit, there are no different forms or versions of each treatment level that could potentially lead to different potential outcomes.

Failure cases: some people get out of date medication

- **Consistency**

$$P(Y=y|X=x, T=t) = P(Y_t=y \mid X=x, T=t)$$

Formalizing Potential Outcomes

- Assume sample of units $i = 1, \dots, n$
- Each has potential outcomes $(Y_1^1, Y_0^1), \dots, (Y_1^n, Y_0^n)$
- The **individual treatment effect (ITE)** for unit i is given by

$$ITE_i = Y_1^i - Y_0^i$$

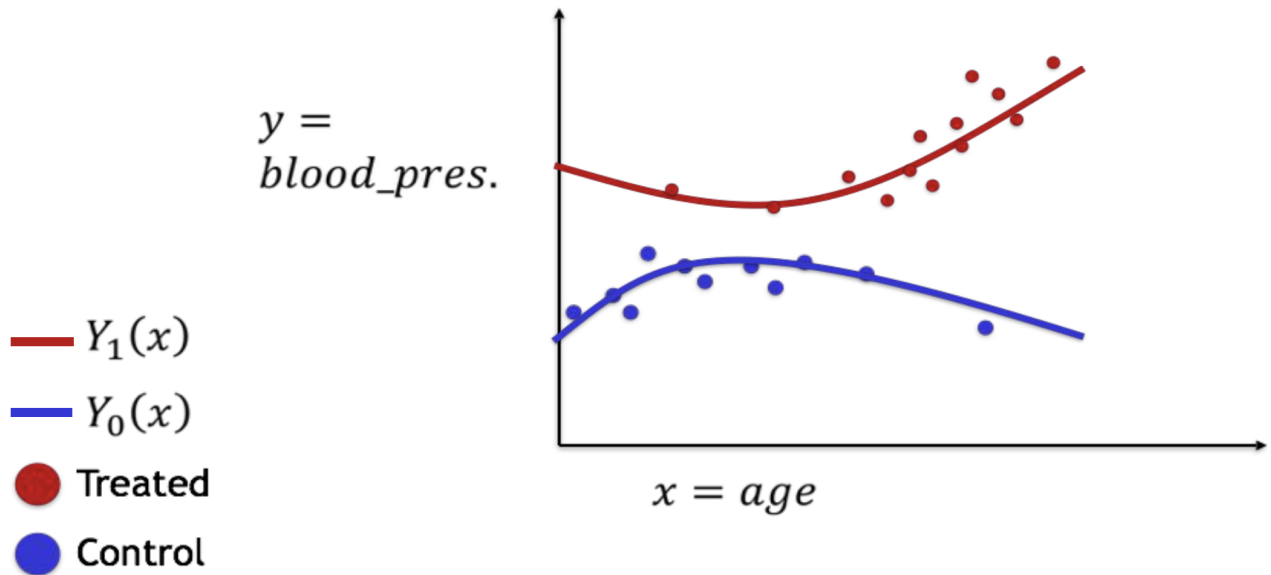
- The **average treatment effect (ATE)** over the sample is:

$$ATE_{\text{finite}} = \frac{1}{n} \sum_{\{i=1\}}^n Y_1^i - Y_0^i$$

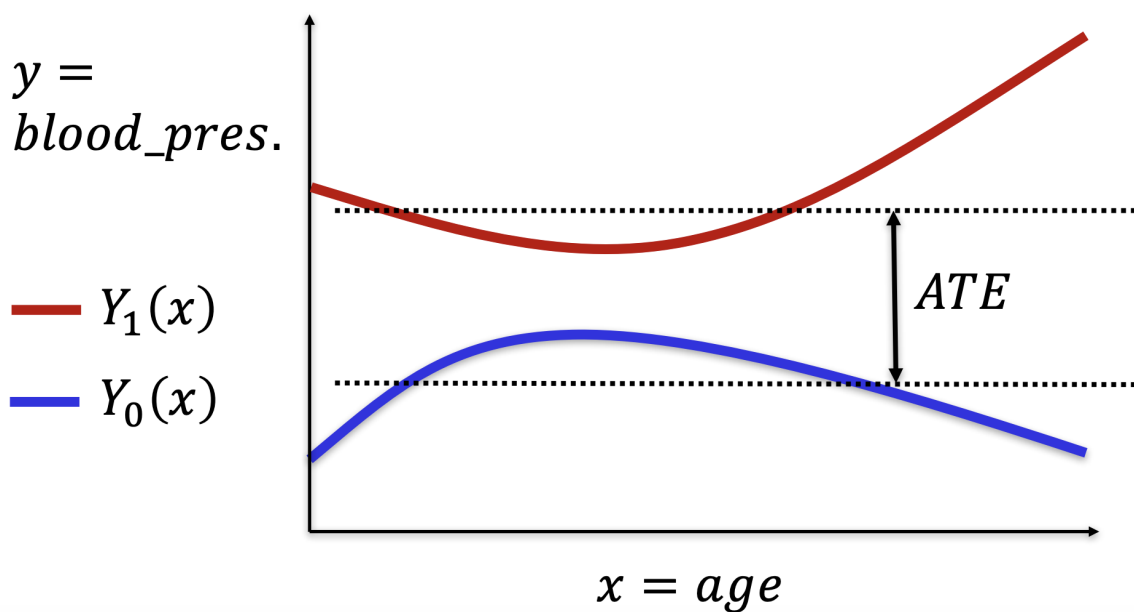
- Assuming some joint distribution $p(Y_0, Y_1)$

$$ATE = \mathbb{E}[Y_1 - Y_0]$$

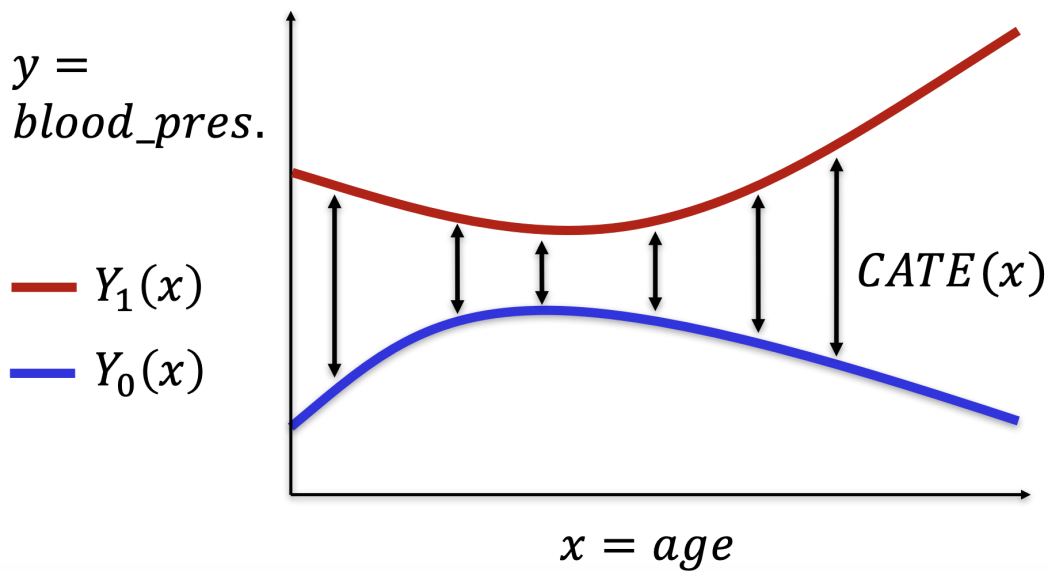
Example: Blood Pressure and Age



Example: Blood Pressure and Age



Example: Blood Pressure and Age



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Example

(age, gender, exercise, treatment)			Observed sugar levels
(45, F, 0, A)			6
(45, F, 1, B)			6.5
(55, M, 0, A)			7
(55, M, 1, B)			8
(65, F, 0, B)			8
(65, F, 1, A)			7.5
(75, M, 0, B)			9
(75, M, 1, A)			8

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Example

(age, gender, exercise)			Observed sugar levels
(45, F, 0)			6
(45, F, 1)			6.5
(55, M, 0)			7
(55, M, 1)			8
(65, F, 0)			8
(65,F, 1)			7.5
(75,M, 0)			9
(75,M, 1)			8

Example

(age, gender, exercise)	Y_0 : Sugar levels <i>had they received medication A</i>	Y_1 : Sugar levels <i>had they received medication B</i>	Observed sugar levels
(45, F, 0)	6	5.5	6
(45, F, 1)	7	6.5	6.5
(55, M, 0)	7	6	7
(55, M, 1)	9	8	8
(65, F, 0)	8.5	8	8
(65,F, 1)	7.5	7	7.5
(75,M, 0)	10	9	9
(75,M, 1)	8	7	8

Example

(age,gender, exercise)	Sugar levels <i>had they received medication A</i>	Sugar levels <i>had they received medication B</i>	Observed sugar levels
(45, F, 0)	6	5.5	6
(45, F, 1)	7	6.5	6.5
(55, M, 0)	7	6	7
(55, M, 1)	9	8	8
(65, F, 0)	8.5	8	8
(65,F, 1)	7.5	7	7.5
(75,M, 0)	10	9	9
(75,M, 1)	8	7	8

$\text{mean}(\text{sugar} \mid \text{medication B}) -$
 $\text{mean}(\text{sugar} \mid \text{medication A}) =$
 ?

$\text{mean}(\text{sugar} \mid \text{had they received B}) -$
 $\text{mean}(\text{sugar} \mid \text{had they received A}) =$
 ?

Example

(age,gender, exercise)	Sugar levels <i>had they received medication A</i>	Sugar levels <i>had they received medication B</i>	Observed sugar levels
(45, F, 0)	6	5.5	6
(45, F, 1)	7	6.5	6.5
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(65,F, 1)	7.5	7	7.5
(75,M, 0)	10	9	9
(75,M, 1)	8	7	8

$\text{mean}(\text{sugar} \mid \text{medication B}) -$
 $\text{mean}(\text{sugar} \mid \text{medication A}) =$
 $7.875 - 7.125 = 0.75$

$\text{mean}(\text{sugar} \mid \text{had they received B}) -$
 $\text{mean}(\text{sugar} \mid \text{had they received A}) =$
 $7.125 - 7.875 = -0.75$

Assumption: Ignorability – no unmeasured confounding

Y_0, Y_1 : potential outcomes for control and treated

x : unit covariates (features)

T : treatment assignment

We assume:

$$(Y_0, Y_1) \perp\!\!\!\perp T \mid x$$

The potential outcomes are independent of treatment assignment, conditioned on covariates x

Assumption: Ignorability – no unmeasured confounding

Y_0, Y_1 : potential outcomes for control and treated

x : unit covariates (features)

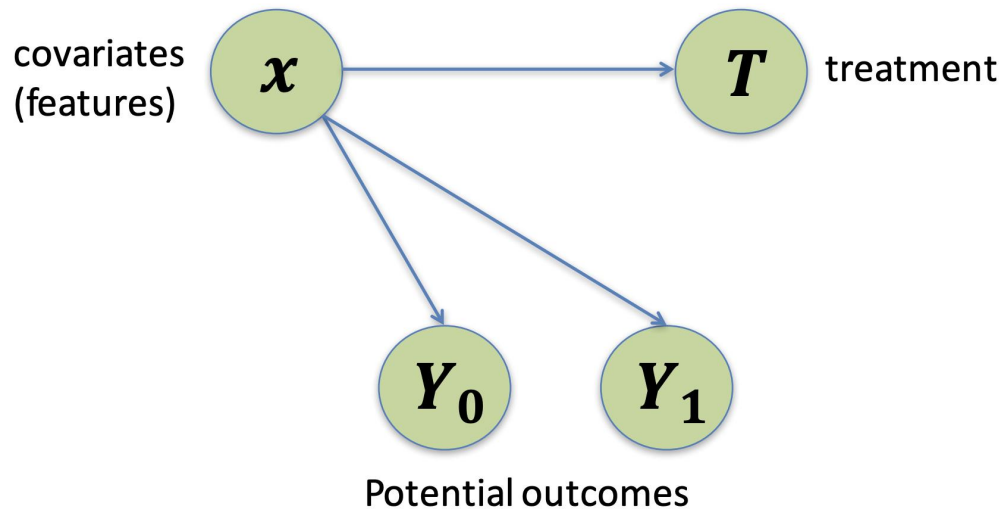
T : treatment assignment

We assume:

$$(Y_0, Y_1) \perp\!\!\!\perp T \mid x$$

Ignorability

Ignorability – no unmeasured confounding

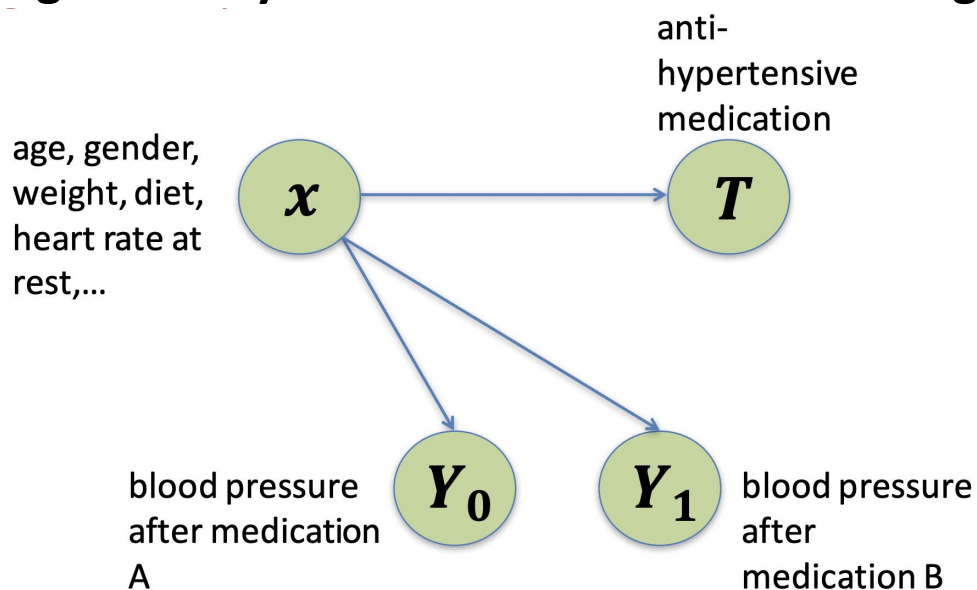


$$(Y_0, Y_1) \perp\!\!\!\perp T \mid x$$

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Ignorability – no unmeasured confounding

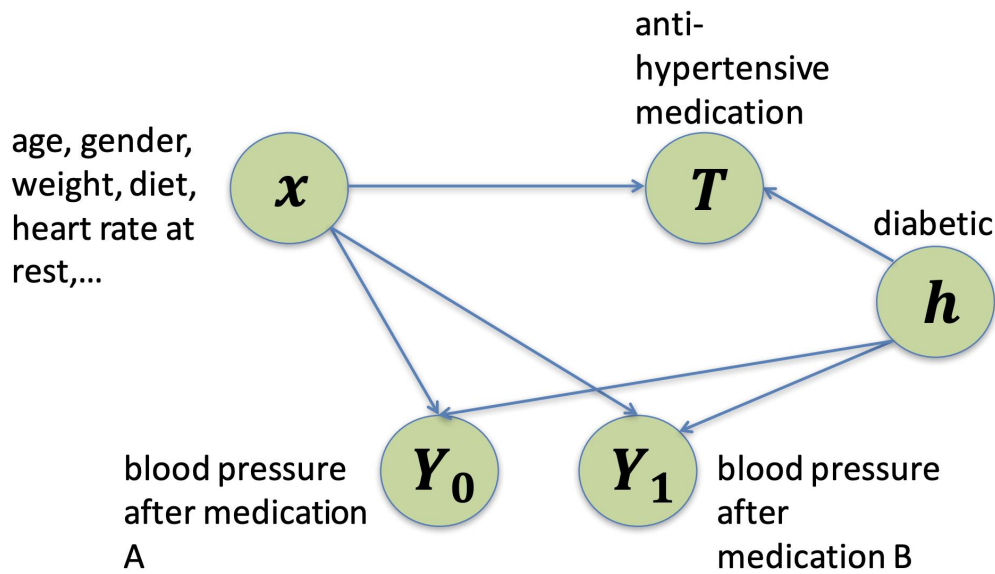


$$(Y_0, Y_1) \perp\!\!\!\perp T \mid x$$

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No ignorability



$$(Y_0, Y_1) \not\perp T \mid x$$

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Assumption: Common Support

Y_0, Y_1 : potential outcomes for control and treated

x : unit covariates (features)

T : treatment assignment

We assume:

$$p(T = t \mid X = x) > 0 \quad \forall t, x$$

Framing the question

1. Where could we go to for data to answer these questions?
2. What should $\mathbf{X}$, T , and Y be to satisfy ignorability?
3. What is the specific causal inference question that we are interested in?
4. Are you worried about common support?

Average Treatment Effect

The expected causal effect of T on Y :

$$ATE := \mathbb{E}[Y_1 - Y_0]$$

Average Treatment Effect

The expected causal effect of T on Y :

$$ATE := \mathbb{E}[Y_1 - Y_0]$$

$$\mathbb{E}[Y_1] =$$

law of total
expectation

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x]] =$$

Average Treatment Effect

The expected causal effect of T on Y :

$$ATE := \mathbb{E}[Y_1 - Y_0]$$

$$\mathbb{E}[Y_1] =$$
$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x]] =$$

ignorability
 $(Y_0, Y_1) \perp\!\!\!\perp T \mid x$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x, T = 1]] =$$

Average Treatment Effect

The expected causal effect of T on Y :

$$ATE := \mathbb{E}[Y_1 - Y_0]$$

$$\mathbb{E}[Y_1] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x]] = \text{ignorability } (Y_0, Y_1) \perp\!\!\!\perp T | x$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x, T = 1]] =$$

Average Treatment Effect

The expected causal effect of T on Y :

$$ATE := \mathbb{E}[Y_1 - Y_0]$$

$$\mathbb{E}[Y_1] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x]] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_1 \sim p(Y_1|x)} [Y_1|x, T = 1]] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}[Y_1|x, T = 1]] \quad \text{shorter notation}$$

Average Treatment Effect

The expected causal effect of T on Y :

$$ATE := \mathbb{E}[Y_1 - Y_0]$$

$$\mathbb{E}[Y_0] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_0 \sim p(Y_0|x)} [Y_0|x]] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}_{Y_0 \sim p(Y_0|x)} [Y_0|x, T = 1]] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}[Y_0|x, T = 0]]$$

The Adjustment Formula

Under the assumption of ignorability, we have that:

$$ATE = \mathbb{E}[Y_1 - Y_0] =$$

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}[Y_1|x, T = 1] - \mathbb{E}[Y_0|x, T = 0]]$$

$$\left. \begin{array}{l} \mathbb{E}[Y_1|x, T = 1] \\ \mathbb{E}[Y_0|x, T = 0] \end{array} \right\} \begin{array}{l} \text{Quantities we} \\ \text{can estimate} \\ \text{from data} \end{array}$$

The Adjustment Formula

Under the assumption of ignorability, we have that:

$$ATE = \mathbb{E}[Y_1 - Y_0] = \mathbb{E}_{x \sim p(x)} [\mathbb{E}[Y_1|x, T=1] - \mathbb{E}[Y_0|x, T=0]]$$

$$\left. \begin{array}{l} \mathbb{E}[Y_0|x, T=1] \\ \mathbb{E}[Y_1|x, T=0] \\ \mathbb{E}[Y_0|x] \\ \mathbb{E}[Y_1|x] \end{array} \right\} \begin{array}{l} \text{Quantities we} \\ \text{cannot directly} \\ \text{estimate from data} \end{array}$$

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The Adjustment Formula

Under the assumption of ignorability, we have that:

$$ATE = \mathbb{E}[Y_1 - Y_0] = \mathbb{E}_{x \sim p(x)} [\mathbb{E}[Y_1|x, T=1] - \mathbb{E}[Y_0|x, T=0]]$$

$$\left. \begin{array}{l} \mathbb{E}[Y_1|x, T=1] \\ \mathbb{E}[Y_0|x, T=0] \end{array} \right\} \begin{array}{l} \text{Quantities we} \\ \text{can estimate} \\ \text{from data} \end{array}$$

Empirically we have samples from $p(x|T=1)$ or $p(x|T=0)$.
Extrapolate to $p(x)$

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Many methods!

Covariate adjustment

Propensity score re-weighting

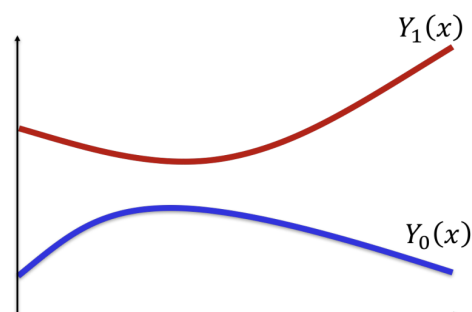
Doubly robust estimators

Matching

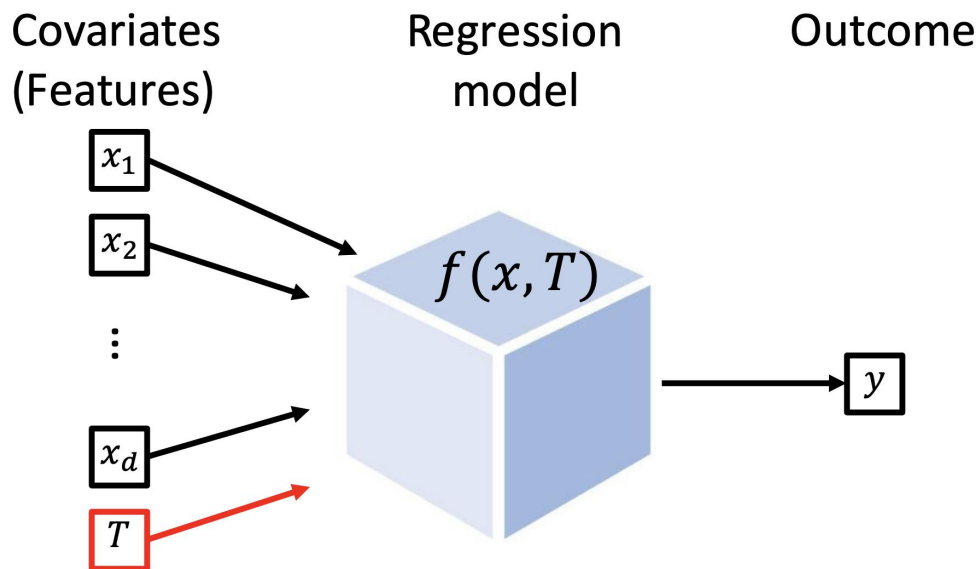
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Covariate adjustment

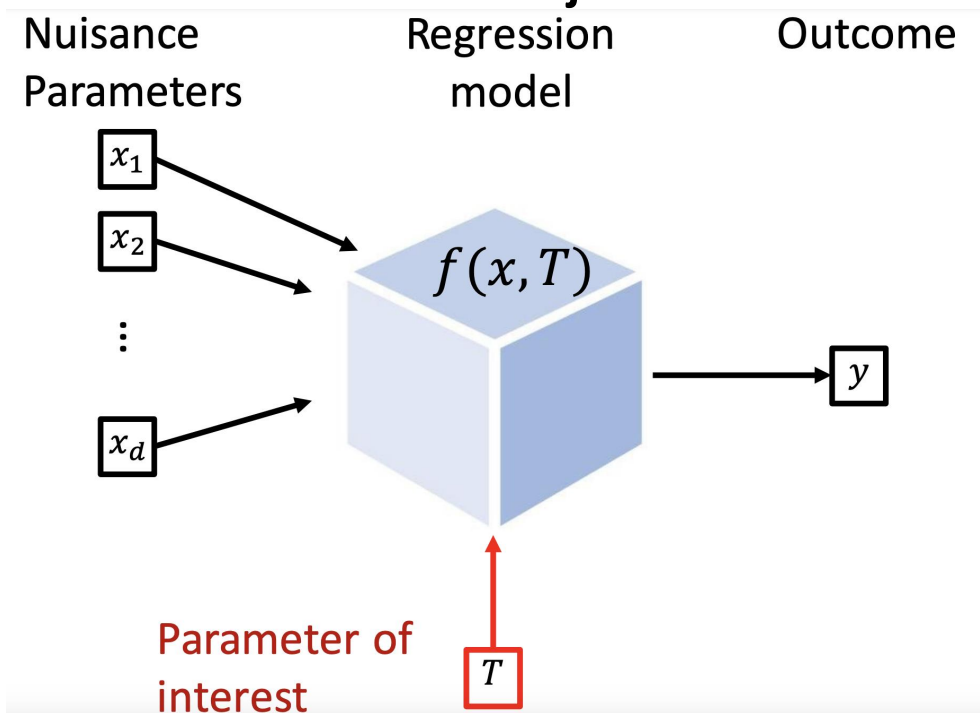
- Explicitly model the relationship between treatment, confounders, and outcome
- Also called “Response Surface Modeling”
- Used for both ITE and ATE
- A regression problem



Covariate adjustment



Covariate adjustment



Covariate adjustment – Parametric g Formula

- Explicitly model the relationship between treatment, confounders, and outcome
- Under ignorability, the expected causal effect of T on Y :

$$\mathbb{E}_{x \sim p(x)} \left[\mathbb{E}[Y_1 | T = 1, x] - \mathbb{E}[Y_0 | T = 0, x] \right]$$

- Fit a model $f(x, t) \approx \mathbb{E}[Y_t | T = t, x]$

$$\widehat{ATE} = \frac{1}{n} \sum_{i=1}^n \left(f(x_i, 1) - f(x_i, 0) \right)$$

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Covariate adjustment – Parametric g Formula

- Explicitly model the relationship between treatment, confounders, and outcome
- Under ignorability, the expected causal effect of T on Y :

$$\mathbb{E}_{x \sim p(x)} \left[\mathbb{E}[Y_1 | T = 1, x] - \mathbb{E}[Y_0 | T = 0, x] \right]$$

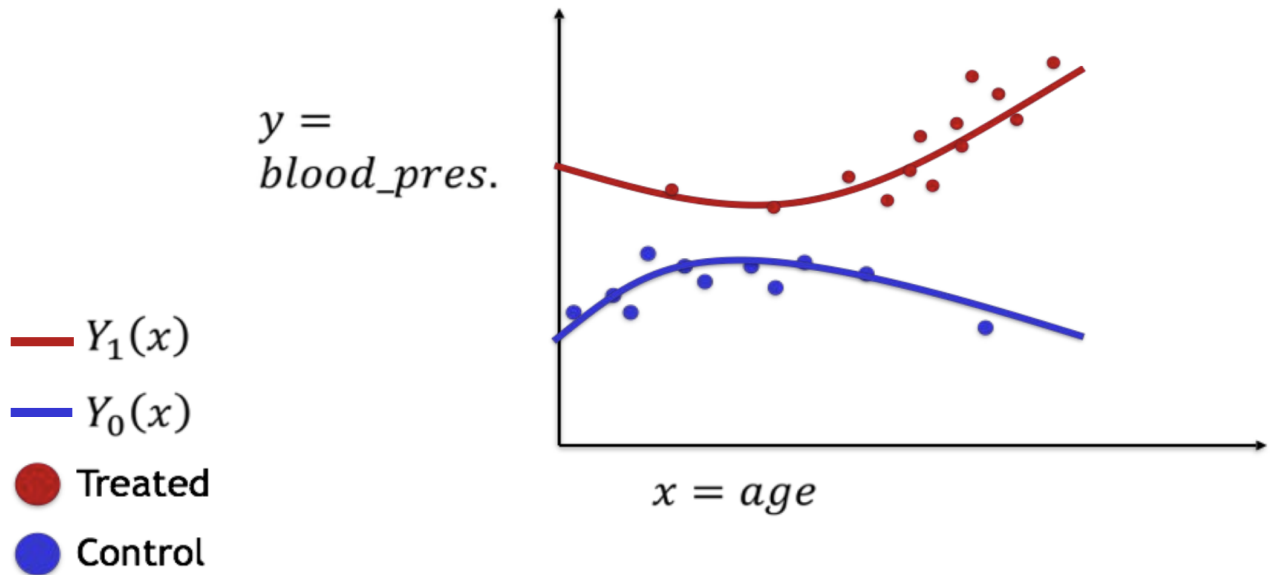
- Fit a model $f(x, t) \approx \mathbb{E}[Y_t | T = t, x]$

$$\widehat{CATE}(x_i) = f(x_i, 1) - f(x_i, 0)$$

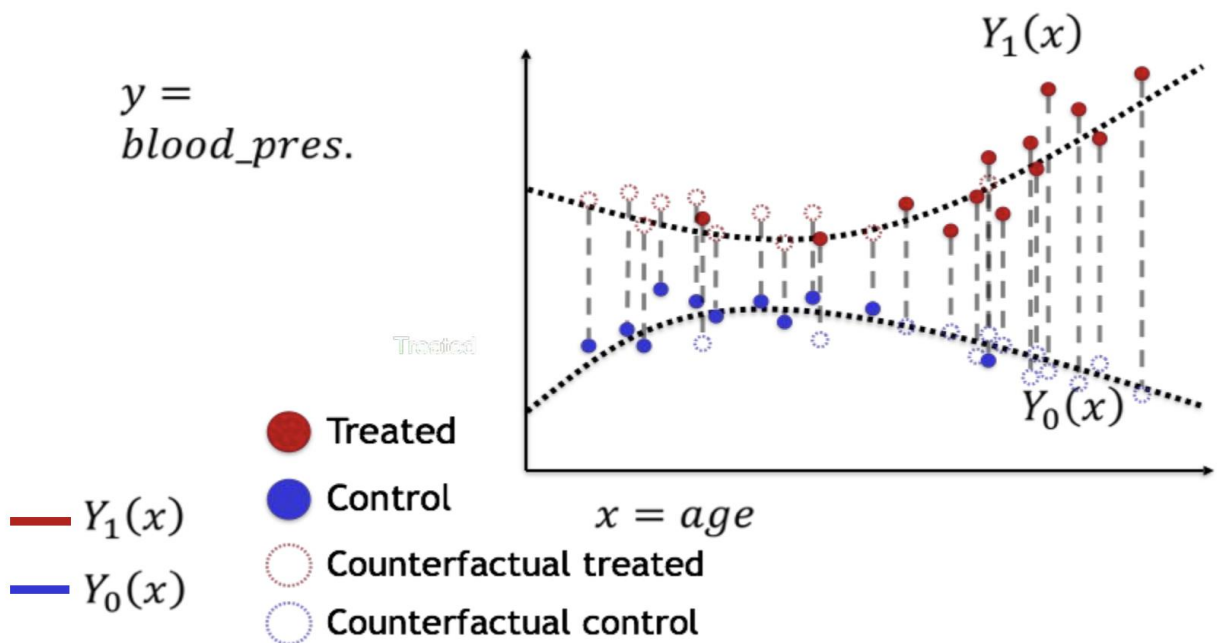
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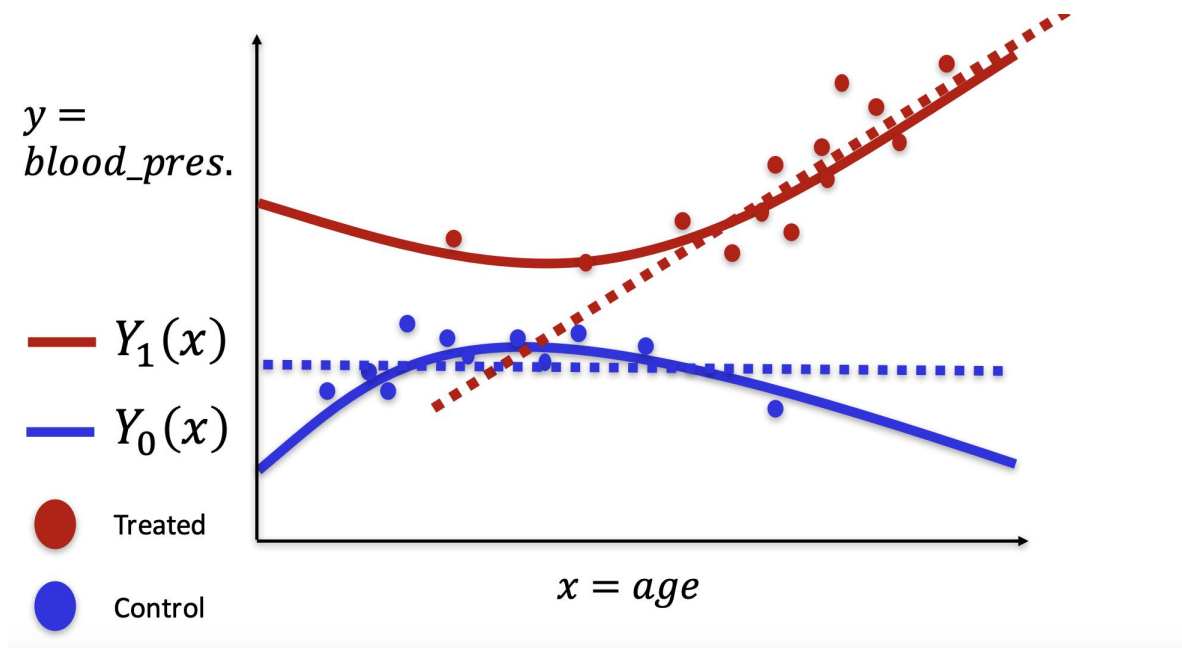
Example: Blood Pressure and Age



Example: Blood Pressure and Age



Example of how adjustment can fail if insufficient overlap



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Covariate adjustment with linear models

- Assume that:

Blood pressure age medication
 $Y_t(x) = \beta x + \gamma \cdot t + \epsilon_t$
 $\mathbb{E}[\epsilon_t] = 0$

- Then:

$$CATE(x) := \mathbb{E}[Y_1(x) - Y_0(x)] =$$

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Covariate adjustment with linear models

- Assume that:

Blood pressure age medication

$$Y_t(x) = \beta x + \gamma \cdot t + \epsilon_t$$

$$\mathbb{E}[\epsilon_t] = 0$$

- Then:

$$CATE(x) := \mathbb{E}[Y_1(x) - Y_0(x)] = \mathbb{E}[(\cancel{\beta x} + \gamma + \epsilon_1) - (\cancel{\beta x} + \epsilon_0)] = \gamma$$

$$ATE := \mathbb{E}_{p(x)}[CATE(x)] = \gamma$$

What happens if the true model is not linear?

- True data generating process, $x \in \mathbb{R}$:

$$Y_t(x) = \beta x + \gamma \cdot t + \delta \cdot x^2$$

$$ATE = \mathbb{E}[Y_1 - Y_0] = \gamma$$

- Hypothesized model:

$$\hat{Y}_t(x) = \hat{\beta}x + \hat{\gamma} \cdot t$$

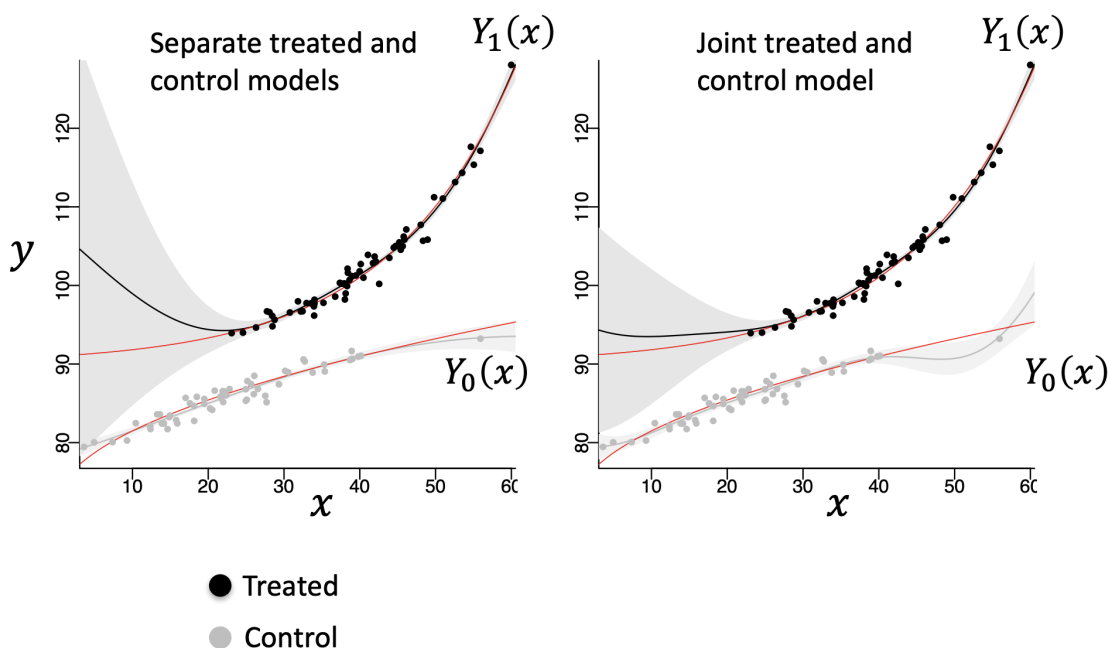
$$\hat{\gamma} = \gamma + \delta \frac{\mathbb{E}[xt]\mathbb{E}[x^2] - \mathbb{E}[t^2]\mathbb{E}[x^2t]}{\mathbb{E}[xt]^2 - \mathbb{E}[x^2]\mathbb{E}[t^2]}$$

Depending on δ , can be made to be arbitrarily large or small!

Covariate adjustment with non-linear models

- Random forests and Bayesian trees
Hill (2011), Athey & Imbens (2015), Wager & Athey (2015)
- Gaussian processes
Hoyer et al. (2009), Zigler et al. (2012)
- Neural networks
Beck et al. (2000), Johansson et al. (2016), Shalit et al. (2016),
Lopez-Paz et al. (2016)

Example: Gaussian Processes



Figures: Vincent Dorie & Jennifer Hill

Example: Neural Networks

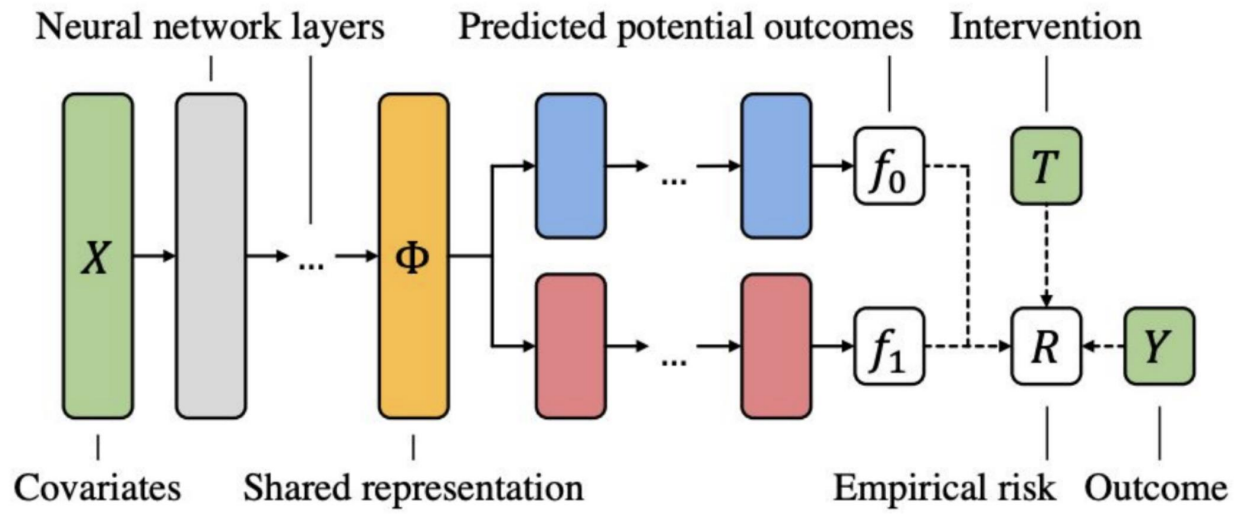
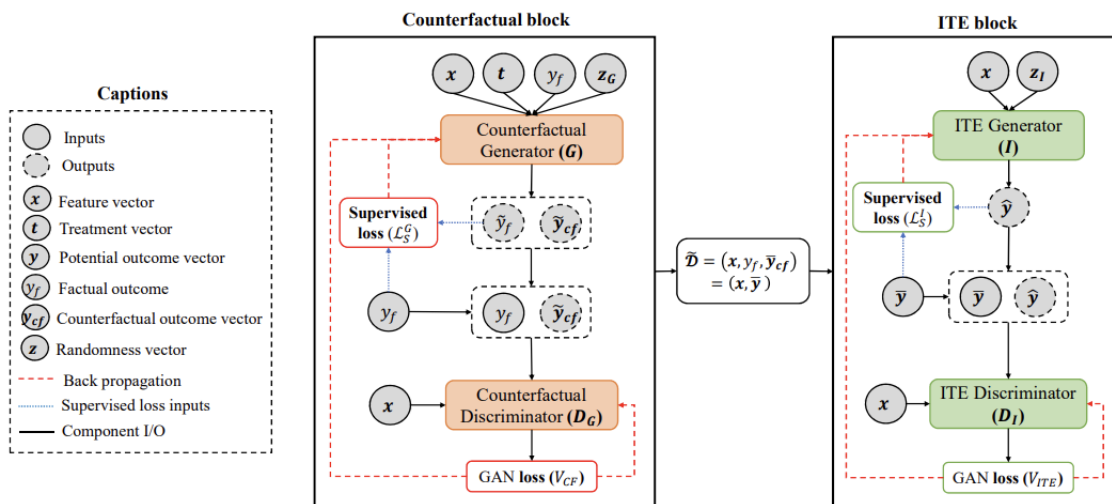


Image from: Johansson et al. "Generalization bounds and representation learning for estimation of potential outcomes and causal effects." *arXiv 2001.07426* (2020).

Example: Neural Networks



Matching

- Find each unit's long-lost counterfactual identical twin, check up on his outcome



Obama, had he gone to law school

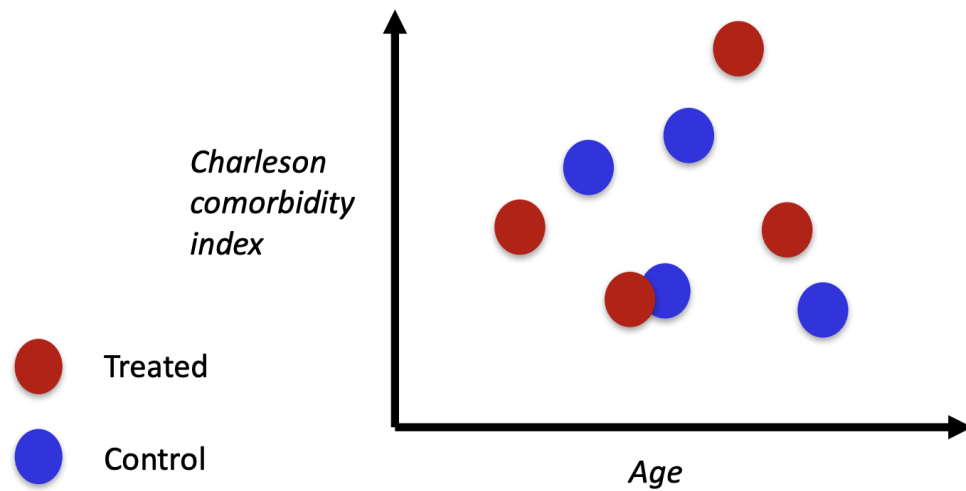


Obama, had he gone to business school

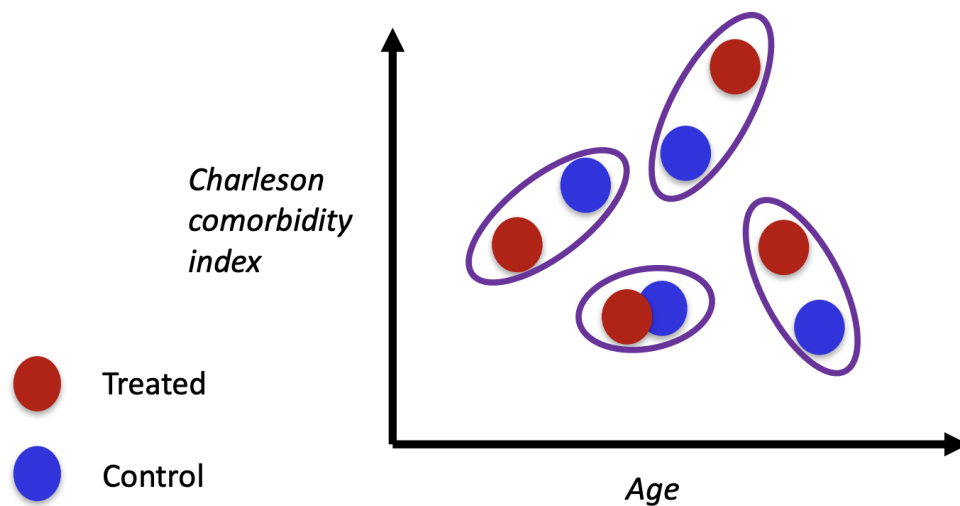
Matching

- Find each unit's long-lost counterfactual identical twin, check up on his outcome
- Used for estimating both ATE and CATE

Match to the nearest neighbour of opposite group



Match to the nearest neighbour of opposite group



1-Nearest Neighbour Matching

- Let $d(\cdot, \cdot)$ be a metric between x 's
- For each i , define $j(i) = \underset{j \text{ s.t. } t_j \neq t_i}{\operatorname{argmin}} d(x_j, x_i)$
 $j(i)$ is the nearest counterfactual neighbor of i

- $t_i = 1$, unit i is treated:

$$\widehat{CATE}(x_i) = y_i - y_{j(i)}$$

- $t_i = 0$, unit i is control:

$$\widehat{CATE}(x_i) = y_{j(i)} - y_i$$

Matching

- Interpretable, especially in small-sample regime
- Nonparametric
- Heavily reliant on the underlying metric
- Could be misled by features which don't affect the outcome

Covariate Adjustment and Matching

- Matching is equivalent to covariate adjustment with two 1-nearest neighbor classifiers:

$$\hat{Y}_1(x) = y_{NN_1(x)}, \hat{Y}_0(x) = y_{NN_0(x)}$$

where $y_{NN_t(x)}$ is the nearest-neighbor of x among units with treatment assignment

$$t = 0,1$$

- 1-NN matching is in general inconsistent, though only with small bias (Imbens 2004)

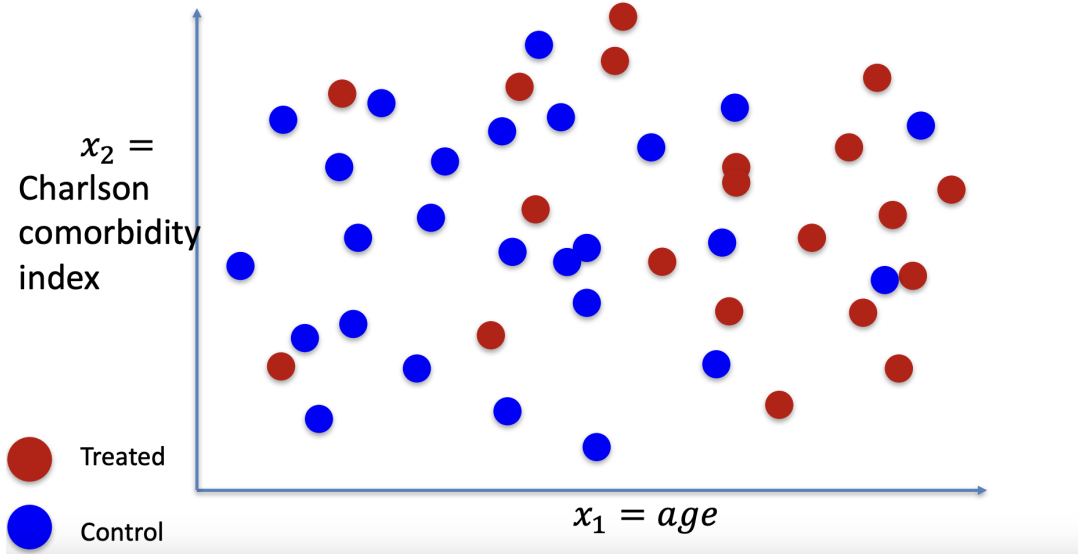
Propensity Scores

- Tool for estimating ATE
- Basic idea: turn observational study into a pseudo-randomized trial by re-weighting samples, similar to importance sampling

Inverse Propensity Reweighting

$$p(x|t=0) \neq p(x|t=1)$$

control *treated*



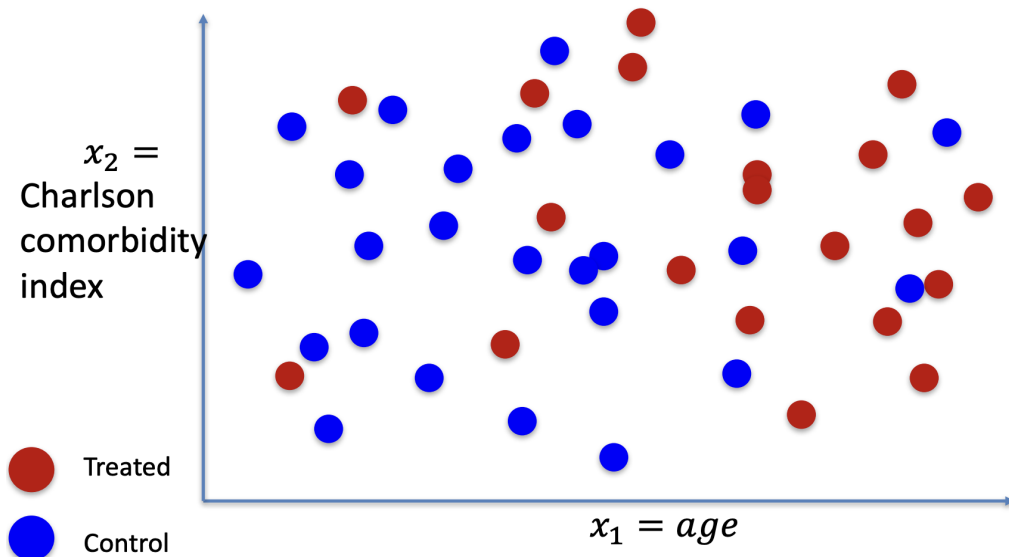
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Inverse Propensity Reweighting

$$p(x|t=0) \cdot w_0(x) \approx p(x|t=1) \cdot w_1(x)$$

reweighted control *reweighted treated*



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Propensity Scores

- Propensity score: $p(T = 1|x)$, using machine learning tools
- Samples re-weighted by the inverse propensity score of the treatment they received

Propensity Scores: Algorithm

Inverse probability of treatment weighted estimator

How to calculate ATE with propensity score
for sample $(x_1, t_1, y_1), \dots, (x_n, t_n, y_n)$

1. Use any ML method to estimate $\hat{p}(T = t|x)$

$$2. \hat{ATE} = \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{\hat{p}(t_i = 1|x_i)} - \frac{1}{n} \sum_{i \text{ s.t. } t_i=0} \frac{y_i}{\hat{p}(t_i = 0|x_i)}$$

Propensity Scores: Algorithm

Inverse probability of treatment weighted estimator

How to calculate ATE with propensity score
for sample $(x_1, t_1, y_1), \dots, (x_n, t_n, y_n)$

1. Randomized trial $p(T = t|x) = 0.5$

$$2. \hat{ATE} = \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{\hat{p}(t_i = 1|x_i)} - \frac{1}{n} \sum_{i \text{ s.t. } t_i=0} \frac{y_i}{\hat{p}(t_i = 0|x_i)}$$

Propensity Scores: Algorithm

Inverse probability of treatment weighted estimator

How to calculate ATE with propensity score
for sample $(x_1, t_1, y_1), \dots, (x_n, t_n, y_n)$

1. Randomized trial $p(T = t|x) = 0.5$

$$2. \hat{ATE} = \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{0.5} - \frac{1}{n} \sum_{i \text{ s.t. } t_i=0} \frac{y_i}{0.5} =$$

Propensity Scores: Algorithm

Inverse probability of treatment weighted estimator

How to calculate ATE with propensity score
for sample $(x_1, t_1, y_1), \dots, (x_n, t_n, y_n)$

1. Randomized trial $p = 0.5$

$$\begin{aligned} 2. \hat{ATE} &= \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{0.5} - \frac{1}{n} \sum_{i \text{ s.t. } t_i=0} \frac{y_i}{0.5} = \\ &= \frac{2}{n} \sum_{i \text{ s.t. } t_i=1} y_i - \frac{2}{n} \sum_{i \text{ s.t. } t_i=0} y_i \end{aligned}$$

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Propensity Scores: Algorithm

- Recall average treatment effect:

$$\mathbb{E}_{x \sim p(x)} [\mathbb{E}[Y_1 | x, T = 1] - \mathbb{E}[Y_0 | x, T = 0]]$$

- We only have samples for:

$$\mathbb{E}_{x \sim p(x|T=1)} [\mathbb{E}[Y_1 | x, T = 1]]$$

$$\mathbb{E}_{x \sim p(x|T=0)} [\mathbb{E}[Y_0 | x, T = 0]]$$

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$$\mathbb{E}_{x \sim p(x|T=1)} [\mathbb{E}[Y_1|x, T=1]]$$

$$\mathbb{E}_{x \sim p(x|T=0)} [\mathbb{E}[Y_0|x, T=0]]$$

- We need to turn $p(x|T=1)$ into $p(x)$:

$$p(x|T=1) \cdot \quad ? \quad = p(x)$$

Propensity Scores: Algorithm

- We only have samples for:

$$\mathbb{E}_{x \sim p(x|T=1)} [\mathbb{E}[Y_1|x, T=1]]$$

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- We need to turn $p(x|T=1)$ into $p(x)$:

$$p(x|T=1) \cdot \frac{p(T=1)}{p(T=1|x)} = p(x)$$

Propensity score

Propensity Scores: Algorithm

- We only have samples for:

$$\mathbb{E}_{x \sim p(x|T=1)} [\mathbb{E}[Y_1|x, T=1]]$$

$$\mathbb{E}_{x \sim p(x|T=0)} [\mathbb{E}[Y_0|x, T=0]]$$

- We need to turn $p(x|T=0)$ into $p(x)$:

$$p(x|T=0) \cdot \frac{p(T=0)}{\underbrace{p(T=0|x)}_{\text{Propensity score}}} = p(x)$$

Propensity Scores: Algorithm

- We want: $\mathbb{E}_{x \sim p(x)} [Y_1(x)]$

- We know that:

$$p(x|T=1) \cdot \frac{p(T=1)}{p(T=1|x)} = p(x)$$

- Thus:

$$\mathbb{E}_{x \sim p(x|T=1)} \left[\frac{p(T=1)}{p(T=1|x)} Y_1(x) \right] = \mathbb{E}_{x \sim p(x)} [Y_1(x)]$$

- We can approximate this empirically as:

$$\frac{1}{n_1} \sum_{i \text{ s.t. } t_i=1} \left[\frac{n_1/n}{\hat{p}(t_i=1|x_i)} y_i \right] = \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{\hat{p}(t_i=1|x_i)}$$

(similarly for $t_i=0$)

Problems with IPW

- Need to estimate propensity score (problem in all propensity score methods)
- If there's not much overlap, propensity scores become non-informative and easily mis-calibrated
- Weighting by inverse can create large variance and large errors for small propensity scores
 - Exacerbated when more than two treatments

Summary

- Two approaches to use machine learning for causal inference:
 1. Predict outcome given features and treatment, then use resulting model to impute counterfactuals (*covariate adjustment*)
 2. Predict treatment using features (*propensity score*), then use to reweight outcome or stratify the data
- Causal graphs important for thinking through whether problem is setup appropriately and whether assumptions hold